

Question:

Alice chooses a set \$A\$ of positive integers. Then Bob lists all finite nonempty sets \$B\$ of positive integers with the property that the maximum element of \$B\$ belongs to \$A\$. Bob's list has 2024 sets. Find the sum of the elements of A.

Reasoning + Answer:

<think>

1. **Understand the problem**:

- Alice picks a set (A) of positive integers .

- Bob creates a list of all finite non empty sets (B) of positive integers such that the maximum element of (B) is in (A) .

- The total number of sets in Bob's list is 2024 .

2. **Analyze the sets (B)** :

- For a given $(a \in A)$, the set (B) can be any subset of $(\{1, 2, 3, \ldots, a-1, a\})$ (excluding (a)).

- The number of subsets of $(\{1, 2, 3, \ldots, a-1\})$ is (2^{a-1}) .

3. **Consider the number of sets (B)** :

- For each $(a \in A)$, the number of sets (B) (where the maximum element is (a)) is (2^{a-1}) .

4. **Sum of (2^{a-1}) for all $(a \in A)$** :

- If $(A = \{a_1, a_2, \ldots, a_n\})$, then the total number of sets (B) is:

$$\sum_{i=1}^n 2^{a_i-1}$$

- According to the problem , this sum equals 2024 .

5. **Integer constraints**:

- Since the number of sets is (2024) , which is (2^{10}) , the sum $(2^{a_1-1} + 2^{a_2-1} + \cdots + 2^{a_n-1} = 2^{10})$.

- This implies $(n = 10)$ because the sum of (10) terms each a power of 2 (including the possibility of some being 1) equals (2^{10}) .

6. **Finding the sum of (a_i)** :

- The set (A) must be $(\{1, 2, 3, \ldots, 10\})$ because $(2^0 + 2^1 + 2^2 + \cdots + 2^9 = 2^{10})$.

7. **Calculate the sum**:

- The sum of (a_i) for $(A = \{1, 2, 3, \ldots, 10\})$ is:

$$1 + 2 + 3 + \cdots + 10$$

- The sum of the first (n) positive integers is given by :

$$\frac{n(n+1)}{2}$$

- For $(n = 10)$:

$$\frac{10 \times 11}{2} = 55$$

</think>

<answer>

55

</answer>